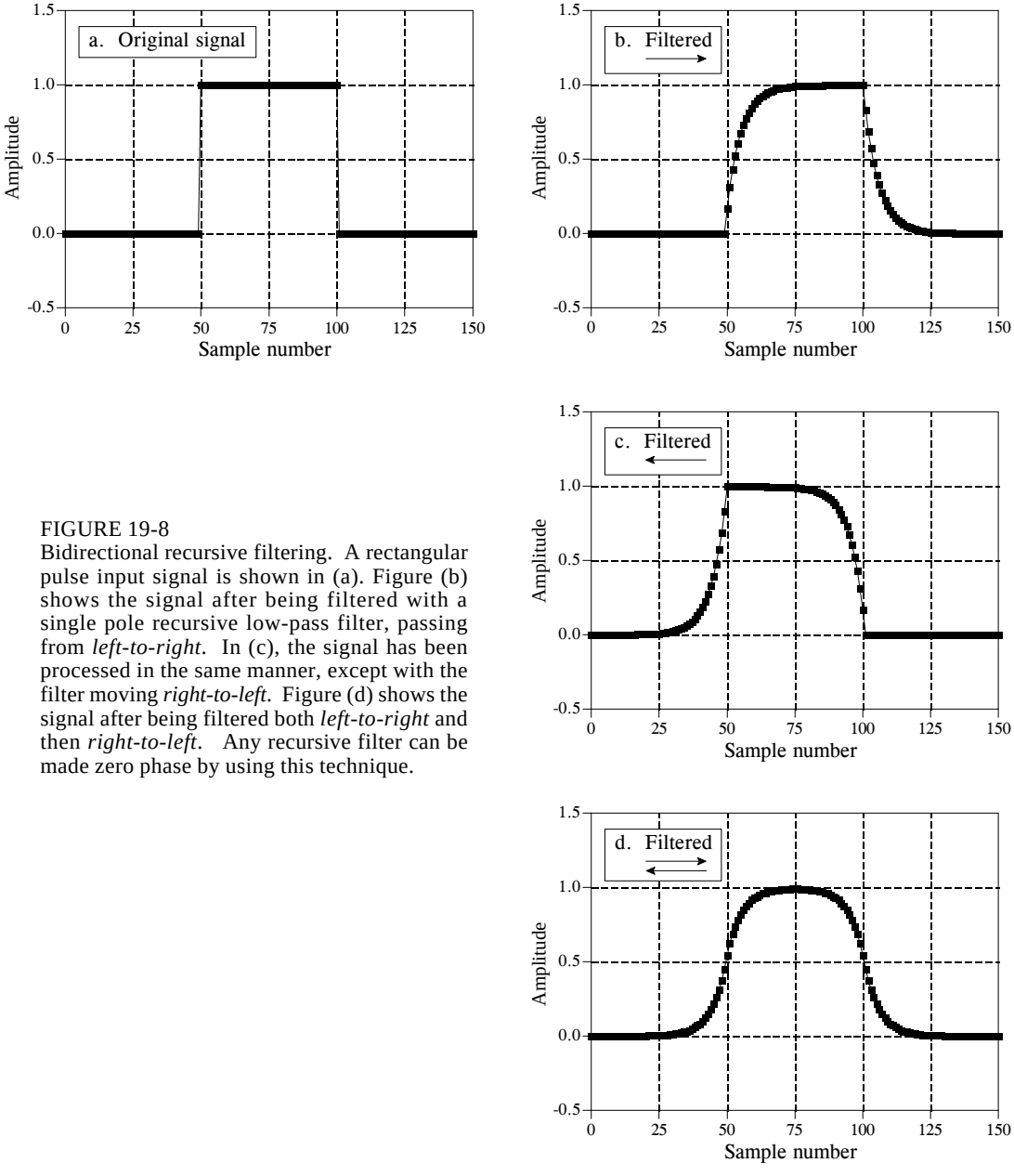

FIGURE 19-8
Bidirectional recursive filtering. A rectangular pulse input signal is shown in (a). Figure (b) shows the signal after being filtered with a single pole recursive low-pass filter, passing from *left-to-right*. In (c), the signal has been processed in the same manner, except with the filter moving *right-to-left*. Figure (d) shows the signal after being filtered both *left-to-right* and then *right-to-left*. Any recursive filter can be made zero phase by using this technique.

How do you find the impulse and frequency responses of the overall filter? The magnitude of the frequency response is the same for each direction, while the phases are opposite in sign. When the two directions are combined, the magnitude becomes *squared*, while the phase cancels to *zero*. In the time domain, this corresponds to convolving the original impulse response with a left-for-right flipped version of itself. For instance, the impulse response of a